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TREND-BASED MOCK PAPER

GATE Computer Science and Information Technology

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Computer Science and Information Technology
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [2-MCQ] [2 marks]

Synonym of 'ubiquitous':

- (A) Rare
- (B) Omnipresent
- (C) Hidden
- (D) Unique

Section B: Computer Science and Information Technology

Q2. [2-MCQ] [2 marks]

LCS uses:

- (A) Greedy
- (B) Divide and Conquer
- (C) Dynamic Programming
- (D) Backtracking

Q3. [2-MCQ] [2 marks]

Merge sort space:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n^2)$

Q4. [2-NAT] [2 marks]

Kruskal uses:

(Numerical Answer Type — Enter the numerical value)

Q5. [2-MCQ] [2 marks]

Time complexity of binary search:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n \log n)$
- (D) $O(1)$

Q6. [2-MCQ] [2 marks]

BFS uses:

- (A) Stack
- (B) Queue
- (C) Heap
- (D) Tree

Q7. [3-MCQ] [4 marks]

The time complexity of Dijkstra's algorithm using a binary heap is:

- (A) $O(V^2)$
- (B) $O(E \log V)$
- (C) $O(V \log V + E)$
- (D) $O(V \log V)$

Q8. [2-NAT] [2 marks]

Worst-case time complexity of merge sort:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

A graph with V vertices and E edges. BFS time complexity:

- (A) $O(V+E)$
- (B) $O(V \cdot E)$
- (C) $O(V \log V)$
- (D) $O(E \log V)$

Q10. [1-MCQ] [1 mark]

Minimum spanning tree with Kruskal's on sorted edges:

- (A) Always unique
- (B) May not be unique
- (C) Never unique
- (D) Depends on graph

Q11. [1-NAT] [1 mark]

Counting sort is stable and runs in:

(Numerical Answer Type — Enter the numerical value)

Q12. [1-MCQ] [1 mark]

Floyd-Warshall computes all-pairs shortest paths in:

- (A) $O(n^3)$
- (B) $O(n^2 \log n)$
- (C) $O(nm)$
- (D) $O(n^2)$

Q13. [1-MCQ] [1 mark]

Strassen matrix multiplication complexity:

- (A) $O(n^3)$
- (B) $O(n^{2.81})$
- (C) $O(n^2 \log n)$
- (D) $O(n \log n)$

Q14. [1-NAT] [1 mark]

A minimum vertex cover in a bipartite graph equals:

(Numerical Answer Type — Enter the numerical value)

Q15. [1-NAT] [1 mark]

Bellman-Ford can detect:

(Numerical Answer Type — Enter the numerical value)

Q16. [1-NAT] [1 mark]

Dijkstra from source s . After extracting u from PQ, which is always true?

(Numerical Answer Type — Enter the numerical value)

Q17. [1-MCQ] [1 mark]

Which algorithm finds k -th smallest in $O(n)$ expected time?

- (A) Quickselect
- (B) Heap select
- (C) Binary search
- (D) Merge select

Q18. [2-NAT] [2 marks]

Round-robin uses:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

Banker's algorithm is for:

- (A) Detection
- (B) Avoidance
- (C) Prevention
- (D) Recovery

Q20. [3-MCQ] [4 marks]

Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2,3,4?

- (A) FIFO
- (B) LRU
- (C) Optimal
- (D) None of these

Q21. [2-NAT] [2 marks]

Inode contains:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

TLB stands for:

- (A) Translation Lookaside Buffer
- (B) Time Local Buffer
- (C) Total Logic Block
- (D) None

Q23. [2-NAT] [2 marks]

Which of the following is NOT a necessary condition for deadlock?

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

Semaphore with value 1 is:

- (A) Counting
- (B) Binary (mutex)
- (C) Monitor
- (D) Lock variable

Q25. [1-MCQ] [1 mark]

Demand paging requires:

- (A) No MMU
- (B) Hardware support (MMU)
- (C) No secondary storage
- (D) Swap space not needed

Q26. [3-MCQ] [3 marks]

3 processes, 3 type-A resources (total). Max needs: P1(2,0,0), P2(2,1,0), P3(1,1,1). Available(1,0,0). Which finishes first?

- (A) P1 only
- (B) P2 only
- (C) P3 only
- (D) P1 and P3

Q27. [2-MCQ] [2 marks]

Derivative of x^3 at $x=2$:

- (A) 6
- (B) 12
- (C) 8
- (D) 4

Q28. [2-NAT] [2 marks]

Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-NAT] [2 marks]

Mean of binomial $n=10, p=0.5$:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

Probability of at least one head in 3 coin tosses:

- (A) $1/8$
- (B) $3/8$
- (C) $7/8$
- (D) $1/2$

Q31. [2-MCQ] [2 marks]

Determinant of 2×2 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) -1

Q32. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q33. [1-NAT] [1 mark]

The Laplace transform of $t \cdot e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q34. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q35. [1-MCQ] [1 mark]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q36. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q37. [2-MCQ] [2 marks]

Asymmetric encryption:

- (A) AES
- (B) DES
- (C) RSA
- (D) 3DES

Q38. [1-NAT] [1 mark]

CSMA/CD is used in:

(Numerical Answer Type — Enter the numerical value)

Q39. [1-MCQ] [1 mark]

Protocol to find MAC from IP:

- (A) DNS
- (B) ARP
- (C) DHCP
- (D) ICMP

Q40. [1-NAT] [1 mark]

Which protocol operates at the application layer of the TCP/IP model?

(Numerical Answer Type — Enter the numerical value)

Q41. [1-MCQ] [1 mark]

TCP connection establishment uses:

- (A) 2-way
- (B) 3-way handshake
- (C) 4-way
- (D) 1-way

Q42. [1-NAT] [1 mark]

OSPF is:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-MCQ] [1 mark]

TCP seq=1000, 500 bytes data. Next expected seq:

- (A) 1000
- (B) 1500
- (C) 500
- (D) 1001

Q44. [1-NAT] [1 mark]

TCP ssthresh initially:

(Numerical Answer Type — Enter the numerical value)

Q45. [1-MCQ] [1 mark]

DHCP is used for:

- (A) Name resolution
- (B) IP address assignment
- (C) Routing
- (D) Email

Q46. [1-MCQ] [1 mark]

Stop-and-Wait utilization with RTT=100ms, $t_{tx}=1ms$:

- (A) 1%
- (B) 1/101
- (C) 99%
- (D) 100%

Q47. [1-MCQ] [1 mark]

In Go-Back-N, sender window size = 7. Sequence numbers needed:

- (A) 7
- (B) 8
- (C) 14
- (D) 6

Q48. [2-MCQ] [2 marks]

Infix to postfix uses:

- (A) Queue
- (B) Stack
- (C) Tree
- (D) Graph

Q49. [2-NAT] [2 marks]

What is the time complexity of inserting at the beginning of a linked list?

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MCQ] [2 marks]

Avg hash lookup:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(1)$
- (D) $O(n^2)$

Q51. [2-MSQ] [2 marks]

Inorder traversal of BST gives:

- (A) Sorted order
- (B) Reverse sorted
- (C) Level order
- (D) Random

Q52. [1-MCQ] [1 mark]

Checkpoint in recovery:

- (A) Forces write
- (B) Creates restore point
- (C) Ends transaction
- (D) Starts log

Q53. [1-NAT] [1 mark]

Schedule: $r_1(X)$, $r_2(X)$, $w_1(X)$, $w_2(X)$, c_1 , c_2 . This is:

(Numerical Answer Type — Enter the numerical value)

Q54. [1-MCQ] [1 mark]

$R(A,B,C,D)$ with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$. Candidate keys:

- (A) $\{A,B\}$, $\{B,C\}$
- (B) $\{A,B\}$, $\{B,D\}$
- (C) $\{B,C\}$, $\{C,D\}$
- (D) $\{A,B\}$ only

Q55. [1-NAT] [1 mark]

Decomposition $R \rightarrow R_1(A,B), R_2(C,D)$ is lossless if:

(Numerical Answer Type — Enter the numerical value)

Q56. [3-MCQ] [3 marks]

Normal form where every non-prime attribute is fully functionally dependent on candidate key:

- (A) 1NF
- (B) 2NF
- (C) 3NF
- (D) BCNF

Q57. [1-MCQ] [1 mark]

The number of Boolean functions of n variables is:

- (A) 2^n
- (B) $2^{(2^n)}$
- (C) $2^{(n^2)}$
- (D) n^2

Q58. [1-NAT] [1 mark]

Full adder has how many inputs:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Flip-flop stores:

- (A) 1 bit
- (B) 2 bits
- (C) 4 bits
- (D) 8 bits

Q60. [1-MCQ] [1 mark]

Halting problem:

- (A) Decidable
- (B) Undecidable
- (C) Semi-decidable
- (D) None

Q61. [1-MCQ] [1 mark]

PDA recognizes:

- (A) Regular
- (B) Context-free
- (C) Context-sensitive
- (D) Recursive

Q62. [1-NAT] [1 mark]

Which language is regular?

(Numerical Answer Type — Enter the numerical value)

Q63. [3-MCQ] [3 marks]

Pipeline with 5 stages at 1 GHz. 10^6 instructions. 30% branches, 2-cycle stall. CPI =

- (A) 1.3

- (B) 1.6
- (C) 1.0
- (D) 2.0

Q64. [1-MCQ] [1 mark]

3-address code operands:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q65. [1-NAT] [1 mark]

Lexical analyzer outputs:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	2MCQ	B	2
2	2MCQ	B	2
3	2NAT	C	2
4	2NAT	B	2
5	2MCQ	C	2
6	2MCQ	B	2
7	2MCQ	B	2
8	2NAT	B	2
9	2MCQ	C	2
10	2MSQ	A	2
11	2MCQ	C	2
12	2MCQ	C	2
13	2NAT	C	2

14	2MCQ	B	2
15	2MCQ	B	2
16	3MCQ	C	4
17	2NAT	B	2
18	2NAT	A	2
19	2MCQ	B	2
20	3MCQ	C	4
21	2NAT	B	2
22	2MCQ	A	2
23	2NAT	C	2
24	2MCQ	B	2
25	2MCQ	C	2
26	1NAT	B	1
27	1MCQ	B	1
28	1NAT	C	1

29	1MCQ	B	1
30	1NAT	B	1
31	1MCQ	B	1
32	1NAT	B	1
33	1MCQ	A	1
34	1MCQ	B	1
35	1MCQ	B	1
36	1NAT	B	1
37	1MCQ	C	1
38	1NAT	A	1
39	1MCQ	B	1
40	1MCQ	B	1
41	1MCQ	A	1
42	1NAT	C	1
43	1NAT	D	1

44	1MCQ	A	1
45	1MCQ	B	1
46	1MCQ	B	1
47	1NAT	A	1
48	1MCQ	B	1
49	1NAT	A	1
50	1NAT	B	1
51	1MCQ	A	1
52	1MCQ	B	1
53	1NAT	B	1
54	1NAT	B	1
55	1MCQ	A	1
56	1NAT	A	1
57	1MCQ	A	1
58	3MCQ	B	3

59	3MCQ	C	3
60	1MCQ	B	1
61	1NAT	A	1
62	3MCQ	B	3
63	1MCQ	B	1
64	1MCQ	A	1
65	1MCQ	B	1

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: difficult

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: moderate

Q3.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: easy

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Probability | Difficulty: difficult

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: difficult

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Programming and Data Structures | Topic: Stacks | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Programming and Data Structures | Topic: Linked List | Difficulty: moderate

Q9.

Answer: C

Solution:

Correct Answer: C

Subject: Programming and Data Structures | Topic: Hashing | Difficulty: easy

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Programming and Data Structures | Topic: Trees | Difficulty: easy

Q11.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Dynamic Programming | Difficulty: moderate

Q12.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Divide and Conquer | Difficulty: moderate

Q13.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Greedy | Difficulty: moderate

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Complexity Analysis | Difficulty: easy

Q15.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Graph Algorithms | Difficulty: easy

Q16.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Graph Algorithms | Difficulty: moderate

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Sorting | Difficulty: moderate

Q18.

Answer: A

Solution:

Correct Answer: A

Subject: Operating Systems | Topic: Process Scheduling | Difficulty: moderate

Q19.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: Deadlocks | Difficulty: easy

Q20.

Answer: C

Solution:

Correct Answer: C

Subject: Operating Systems | Topic: Memory Management | Difficulty: difficult

Q21.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: File Systems | Difficulty: moderate

Q22.

Answer: A

Solution:

Correct Answer: A

Subject: Operating Systems | Topic: Virtual Memory | Difficulty: moderate

Q23.

Answer: C

Solution:

Correct Answer: C

Subject: Operating Systems | Topic: Deadlocks | Difficulty: difficult

Q24.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: Synchronization | Difficulty: moderate

Q25.

Answer: C

Solution:

Correct Answer: C

Subject: Computer Networks | Topic: Security | Difficulty: easy

Q26.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Data Link Layer | Difficulty: easy

Q27.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Network Layer | Difficulty: moderate

Q28.

Answer: C

Solution:

Correct Answer: C

Subject: Computer Networks | Topic: Application Layer | Difficulty: difficult

Q29.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Transport Layer | Difficulty: moderate

Q30.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Routing | Difficulty: difficult

Q31.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Logic | Topic: Boolean Algebra | Difficulty: difficult

Q32.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Logic | Topic: Combinational Circuits | Difficulty: easy

Q33.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Logic | Topic: Sequential Circuits | Difficulty: easy

Q34.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Decidability | Difficulty: moderate

Q35.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Context Free Languages | Difficulty: difficult

Q36.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Regular Languages | Difficulty: easy

Q37.

Answer: C

Solution:

Correct Answer: C

Subject: Compiler Design | Topic: Code Generation | Difficulty: moderate

Q38.

Answer: A

Solution:

Correct Answer: A

Subject: Compiler Design | Topic: Lexical Analysis | Difficulty: easy

Q39.

Answer: B

Solution:

Checkpoint: writes all modified buffers to disk, logs checkpoint entry. Recovery starts from last checkpoint.

Subject: DBMS | Topic: DBMS | Difficulty: moderate

Q40.

Answer: B

Solution:

Next expected = $1000 + 500 = 1500$.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: difficult

Q41.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: C

Solution:

Initial ssthresh = 65536 bytes (64 KB). cwnd starts at 1-4 MSS.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

Q43.

Answer: D

Solution:

T2 reads dirty data (w1 before c1). If T1 aborts, T2 has read uncommitted. Not recoverable.

Subject: DBMS | Topic: DBMS | Difficulty: moderate

Q44.

Answer: A

Solution:

BFS: visits each vertex and edge once. $O(V+E)$ with adjacency list.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q45.

Answer: B

Solution:

Closures: $(AB)^+=ABCD$, $(BD)^+=ABCD$, $(BC)^+=ABCD$. Minimal: AB and BD.

Subject: DBMS | Topic: DBMS | Difficulty: easy

Q46.

Answer: B

Solution:

If edge weights are equal, multiple MSTs possible. Kruskal picks arbitrarily among equal-weight edges.

Subject: Algorithms | Topic: Algorithms | Difficulty: easy

Q47.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: B

Solution:

Demand paging: MMU provides valid-invalid bits. Page fault on invalid access.

Subject: Operating Systems | Topic: Operating Systems | Difficulty: difficult

Q49.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q50.

Answer: B

Solution:

Counting sort: $O(n+k)$ time, $O(k)$ space. Stable with prefix sums.

Subject: Algorithms | Topic: Algorithms | Difficulty: easy

Q51.

Answer: A

Solution:

Floyd-Warshall: $O(n^3)$ DP over all intermediate vertices.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q52.

Answer: B

Solution:

$T(n) = 7T(n/2) + O(n^2)$. Master theorem: $O(n^{\log_2(7)})$ approx $O(n^{2.81})$.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q53.

Answer: B

Solution:

Koenig's theorem: min vertex cover size = max matching size in bipartite graphs.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q54.

Answer: B

Solution:

Bellman-Ford: after $V-1$ relaxations, one more detects negative cycles.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q55.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

Dijkstra extracts in non-decreasing distance order. Once extracted, $d[u]$ is finalized (non-negative edges).

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q57.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q58.

Answer: B

Solution:

Branch penalty: $0.3 * 2 = 0.6$ extra cycles. $CPI = 1 + 0.6 = 1.6$.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: moderate

Q59.

Answer: C

Solution:

Check each: P1 needs 2A (1 available, no). P2 needs 2A (1, no). P3 needs 1A (1 available, needs B=1, 0 available, no). Hmm none can finish. Let me recheck: P3 needs A=1, B=1. Available A=1, B=0. Can't finish. System is in deadlock?

Subject: Operating Systems | Topic: Operating Systems | Difficulty: difficult

Q60.

Answer: B

Solution:

DHCP: Dynamic Host Configuration Protocol. Assigns IP addresses dynamically.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

Q61.

Answer: A

Solution:

Lossless iff $(R1 \cap R2)$ functionally determines R1 or R2.

Subject: DBMS | Topic: DBMS | Difficulty: difficult

Q62.

Answer: B

Solution:

2NF: no partial dependency of non-prime on candidate key. Requires 1NF first.

Subject: DBMS | Topic: DBMS | Difficulty: moderate

Q63.

Answer: B

Solution:

Utilization = $t_{tx} / (t_{tx} + RTT) = 1/101$.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: easy

Q64.

Answer: A

Solution:

Quickselect (selection algorithm): $O(n)$ expected using partition like quicksort.

Subject: Algorithms | Topic: Algorithms | Difficulty: easy

Q65.

Answer: B

Solution:

Window = N: need N+1 sequence numbers to distinguish. For N=7: need 8 (3 bits).

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Algorithms: 16 questions, 25 marks
Operating Systems: 9 questions, 20 marks
Engineering Mathematics: 10 questions, 15 marks
Computer Networks: 11 questions, 12 marks
Programming and Data Structures: 4 questions, 8 marks
DBMS: 5 questions, 7 marks
Digital Logic: 3 questions, 3 marks
Theory of Computation: 3 questions, 3 marks
Computer Organization: 1 questions, 3 marks
General Aptitude: 1 questions, 2 marks
Compiler Design: 2 questions, 2 marks

Difficulty Distribution:

Easy: 18 | Moderate: 34 | Difficult: 13

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.